SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Technical Presentation

Code of Conduct:

I R. Sridhu (192524332) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

R. Sidhu

Signature of the student:

Annexure A

Technical Presentation

Question 1 (5 Marks)
Suppose an Operating System is used in a banking system handling thousands of transactions simultaneously.
Prepare a technical presentation explaining:
Types of OS suitable for this scenario
OS structure used
Role of system calls in transaction processing

Question 2 (5 Marks)
Consider a multi-user Linux-based server environment supporting file sharing and device access.
Prepare a technical presentation covering:
OS functions involved in resource management
File and device system calls
Interaction between user processes and kernel

Question 3 (5 Marks)
Assume a real-time embedded system used in traffic signal control.
Prepare a technical presentation explaining:
Type of OS used and justification
OS structure and components
System calls used for process control and timing

Question 4 (5 Marks)
A company uses a distributed computing system across multiple locations.
Prepare a technical presentation discussing:
Type of OS and its characteristics
Architecture of distributed OS
System calls used for communication and coordination

Question 5 (5 Marks)
Consider an OS managing process scheduling, memory allocation, and I/O operations in a modern computer.
Prepare a technical presentation including:
Key OS functions and their importance
Interaction between OS components
Examples of system calls for each function

RUBRICS FOR CALCULATION

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total MarkS
Q1						
Q2						
Q3						
Q4						
Q5						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

SIMATS ENGINEERING

ASSESSMENT TOOL 3 (AT-3)

OPERATING SYSTEMS

TECHNICAL PRESENTATION

CASE STUDY: BANKING SYSTEM HANDLING THOUSANDS OF TRANSACTIONS SIMULTANEOUSLY

COURSE NAME	COURSE CODE	COURSE OUTCOME	ASSESSMENT TOOL
CSA04	Operating Systems	CO1	AT-3

PRESENTED BY

Name: R. Indhu | Reg No: 192524332

Operating System for Banking Transactions

❖ Banking systems process thousands of transactions every second.

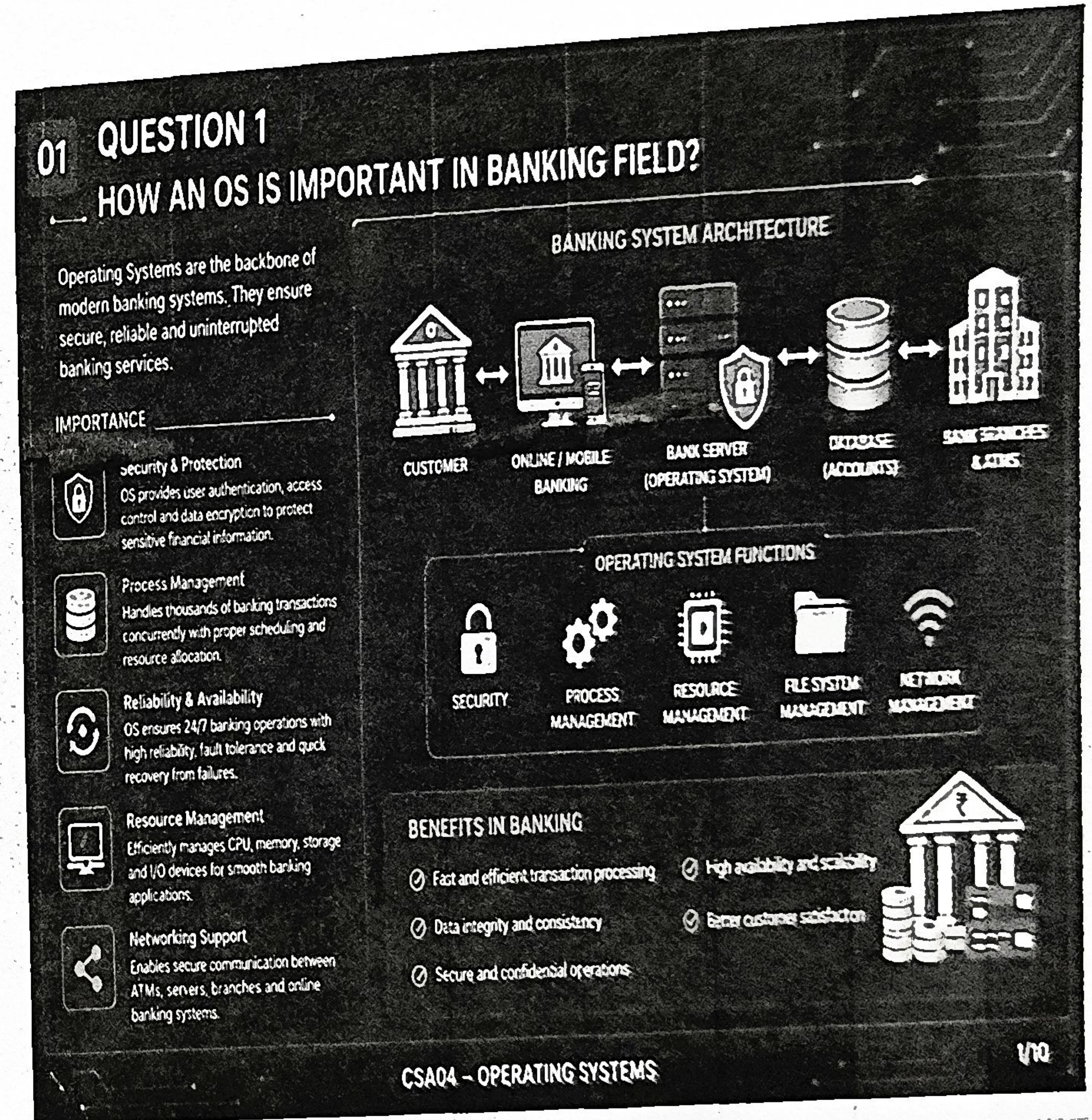
❖ **Suitable OS:** Multi-user Operating System Supports multiple users simultaneously.

❖ Ensures secure and fast transaction processing.

❖ Provides process scheduling and memory management.

❖ Maintains data consistency and reliability.

❖ Prevents system failure during heavy workload.



OS Structure and System Calls

System Calls Used

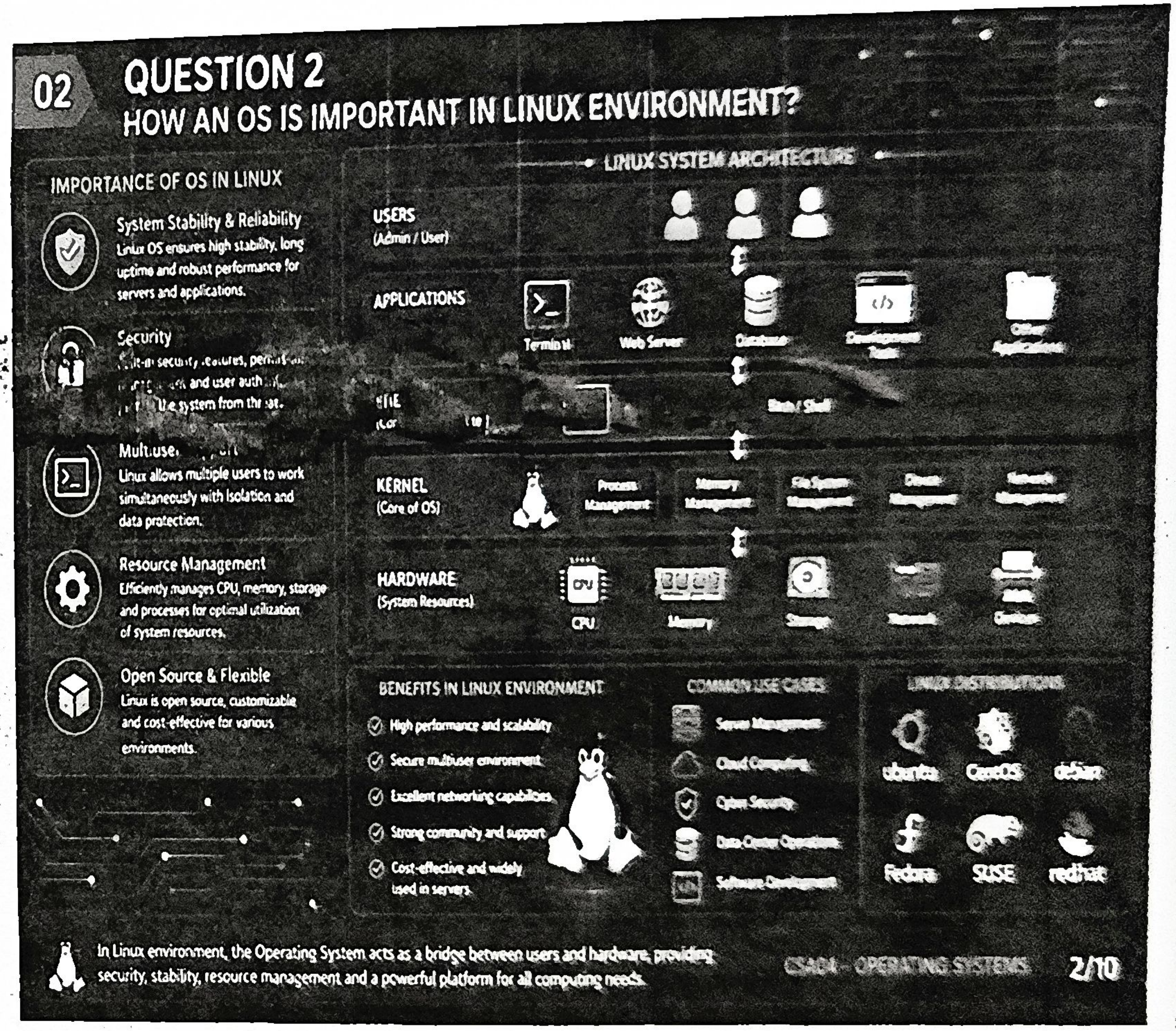
- Process creation (fork())
- File operations (open(), read(), write())
- Memory allocation
- Process synchronization
- Secure transaction handling

OS Structure

- Layered Operating System Structure
- Hardware at the bottom
- Kernel manages system resources
- User applications at the top

Linux Resource Management

- ❖ Linux supports multiple users simultaneously.
- ❖ CPU scheduling for all users.
- ❖ Memory allocation for running programs.
- ❖ File management with permissions.
- ❖ Device management using drivers.
- ❖ Security through user authentication.



System Calls in Linux

File System Calls

open()

read()

write()

close()

Device System Calls

ioctl()

read()

write()

User Process & Kernel

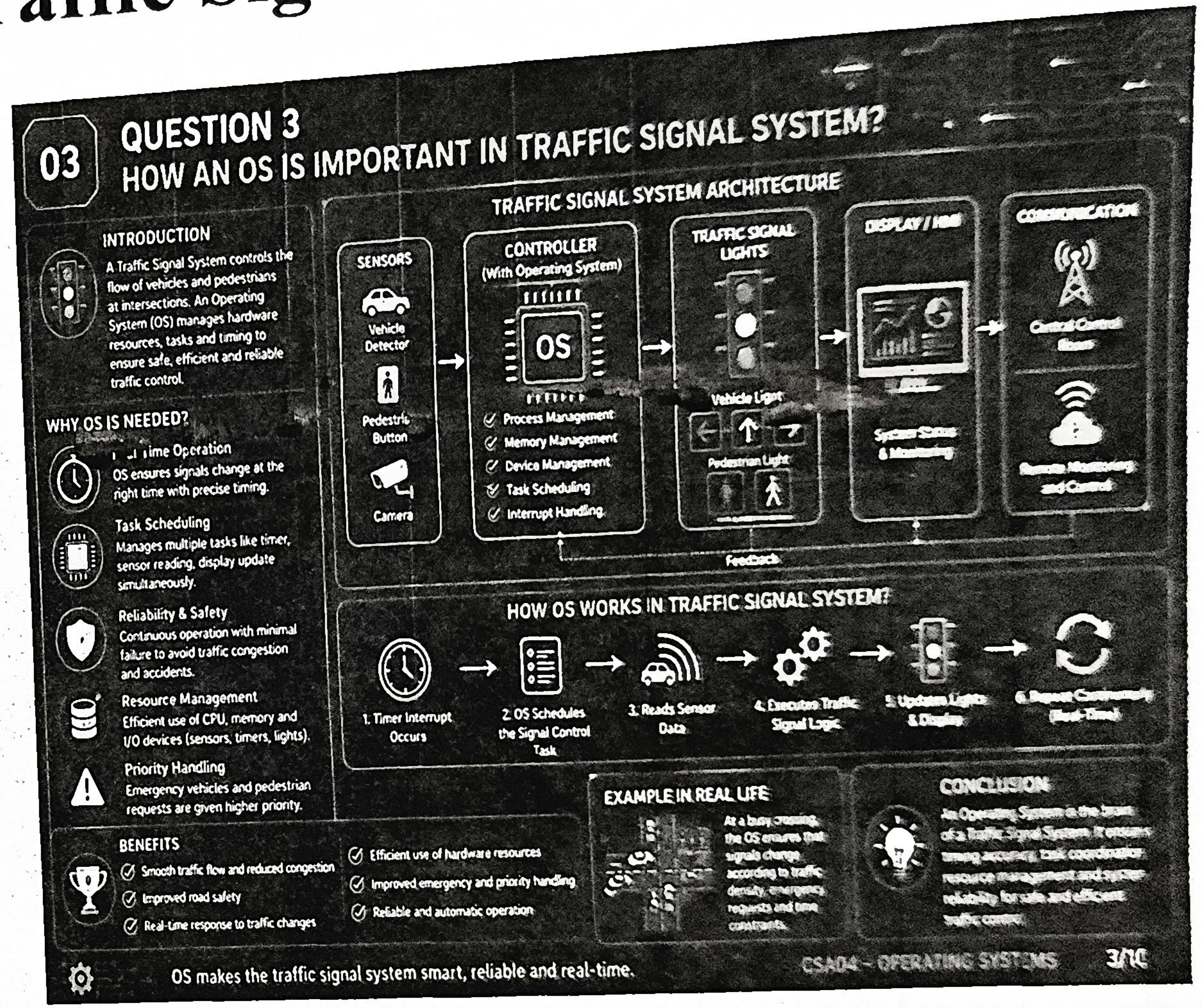
User sends request.

Kernel processes request.

Kernel returns result safely

RTOS in Traffic Signal Control

- ❖ **Operating System:** Real-Time Operating System (RTOS)
- ❖ Gives immediate response.
- ❖ Executes tasks within fixed time.
- ❖ High reliability.
- ❖ Controls traffic lights accurately.
- ❖ Avoids signal delays.



Structure & System Calls

OS Components

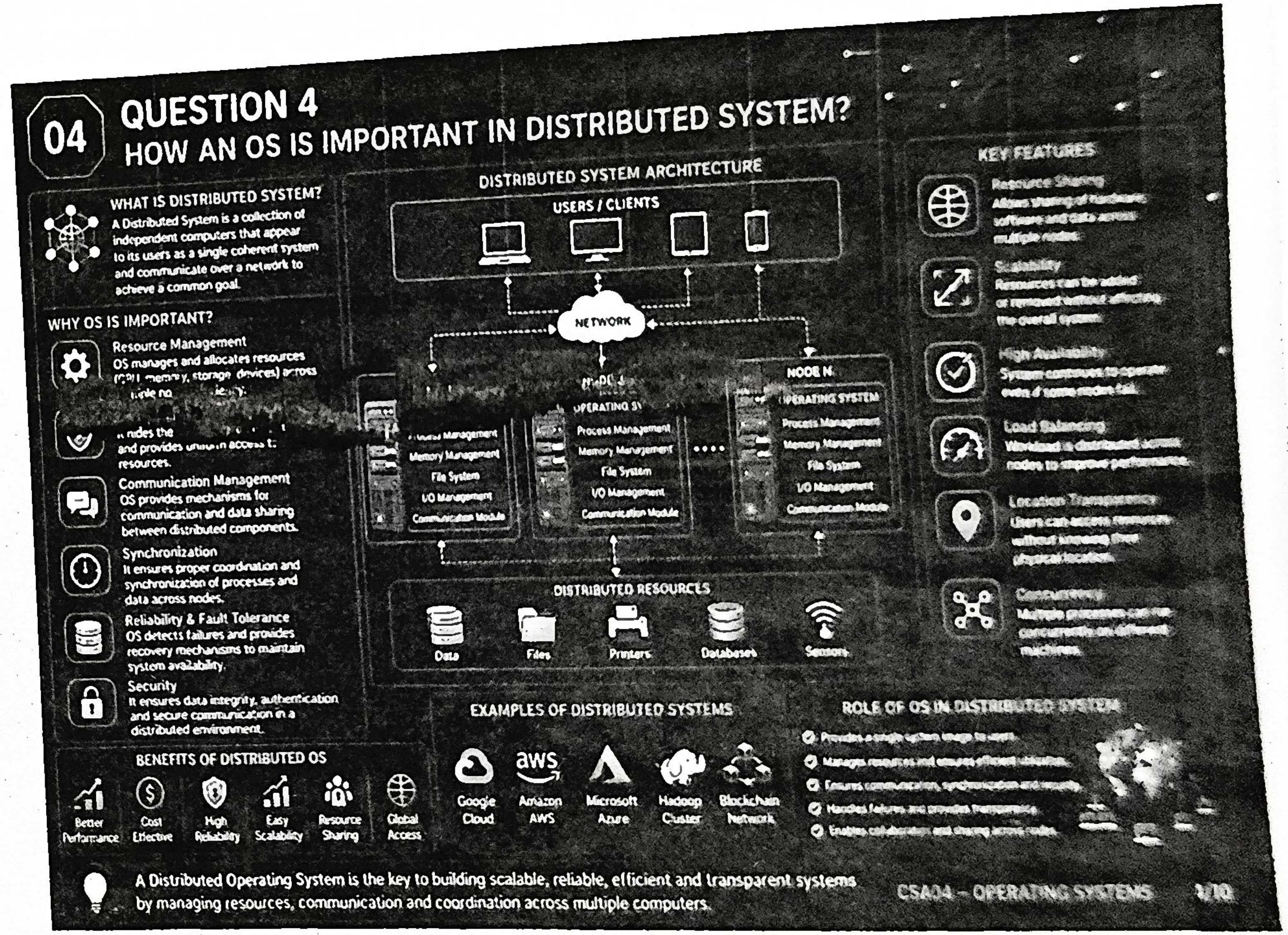
- Kernel
- Scheduler
- Timer
- Interrupt Handler
- Device Drivers

System Calls

- Process creation
- Delay/Timer
- Interrupt handling
- Task synchronization
- Process termination

Distributed Operating System

- ❖ Used across multiple locations.
- ❖ Computers work together as one system.
- ❖ Shares resources efficiently.
- ❖ Improves performance.
- ❖ High availability. Fault tolerance.



Distributed OS Architecture

Architecture

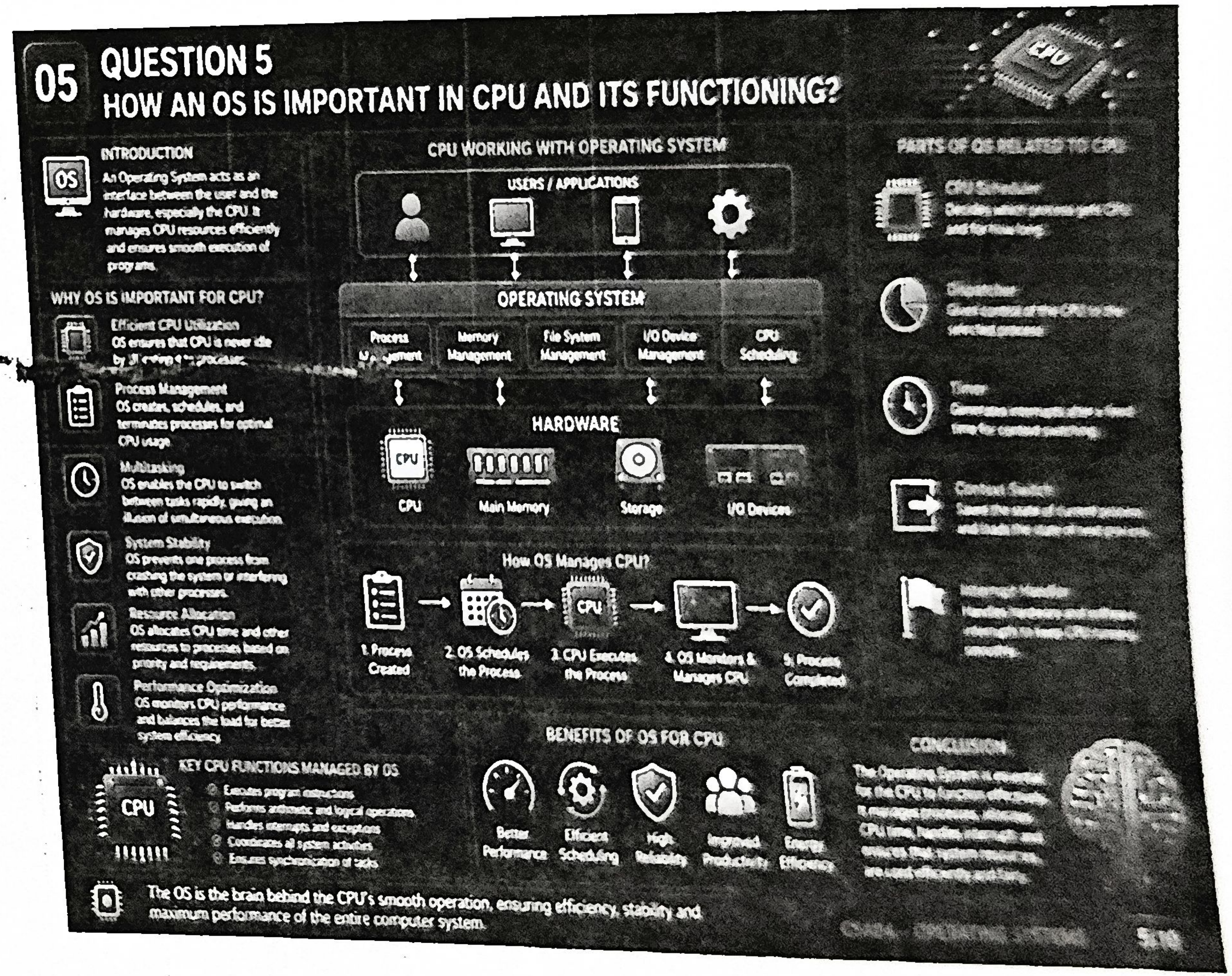
- Client Nodes
- Server Nodes
- Communication Network
- Distributed Kernel

System Calls

- send()
- receive()
- Remote Procedure Call (RPC)
- Socket communication
- Process synchronization

Modern Computer Operating System

- Process scheduling
- Memory management
- File management
- I/O management
- Security management
- Resource allocation
- **Importance**
- Improves system performance.
- Ensures efficient resource usage.
- Provides secure computing.



OS Components & System

System Call Examples

Process → fork(), exec()

Memory → mmap()

File → open(), read(), write()

I/O → ioctl()

Component Interaction

CPU Scheduler manages processes.

Memory Manager allocates RAM.

I/O Manager controls devices.

File System stores data.